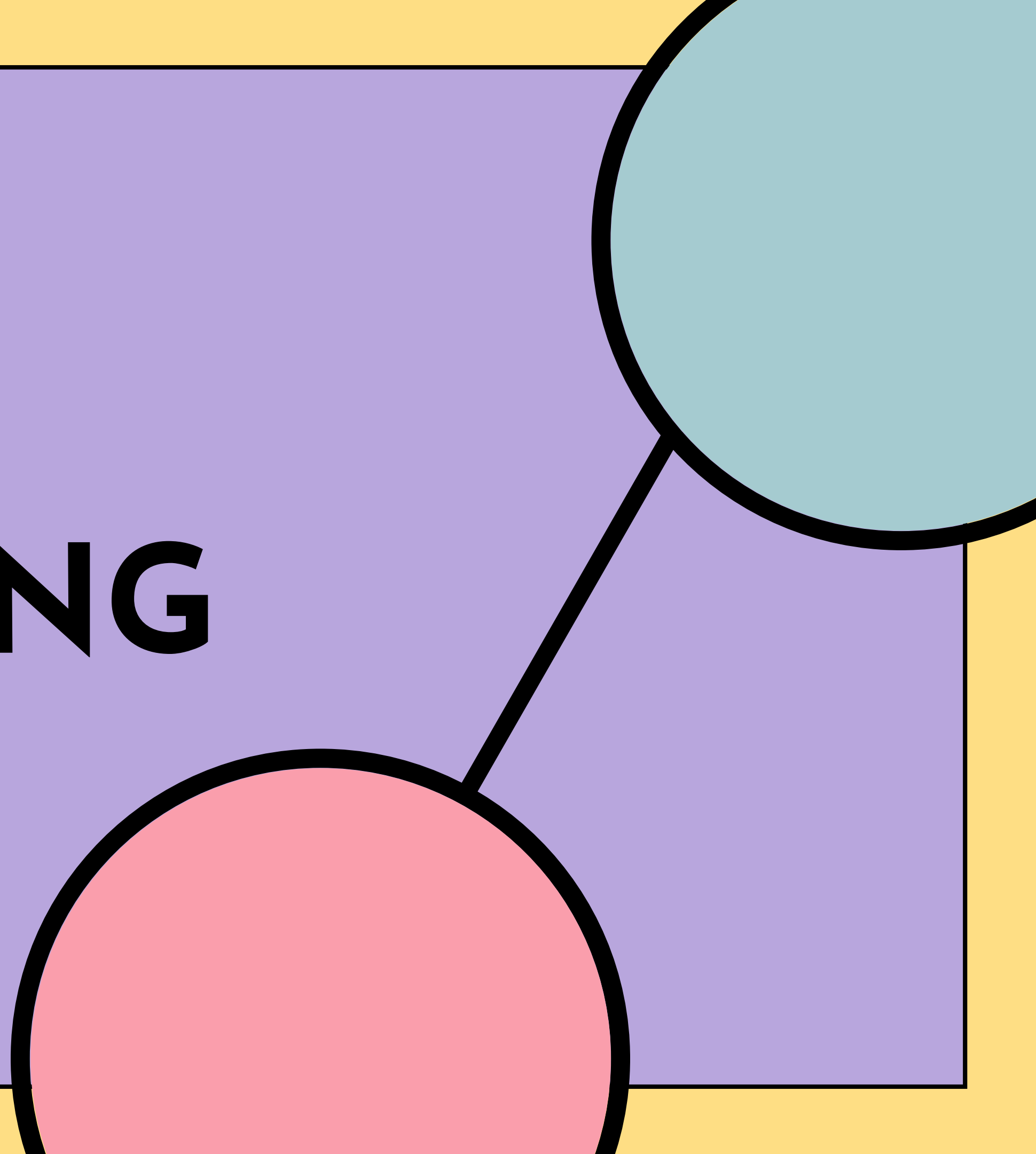


K-MEANS CLUSTERING



Introduction

What is the difference between
Clustering and **Classification**?

~	Classification	Clustering
Input Data	Labeled	Unlabeled
Goal	Predict known labels	Discover hidden patterns
Categories	Number of categories is known	Number of groups is unknown
Evaluation	Accuracy, precision, recall	NO ACCURACY

Instead of **accuracy** we use **cluster quality metrics**

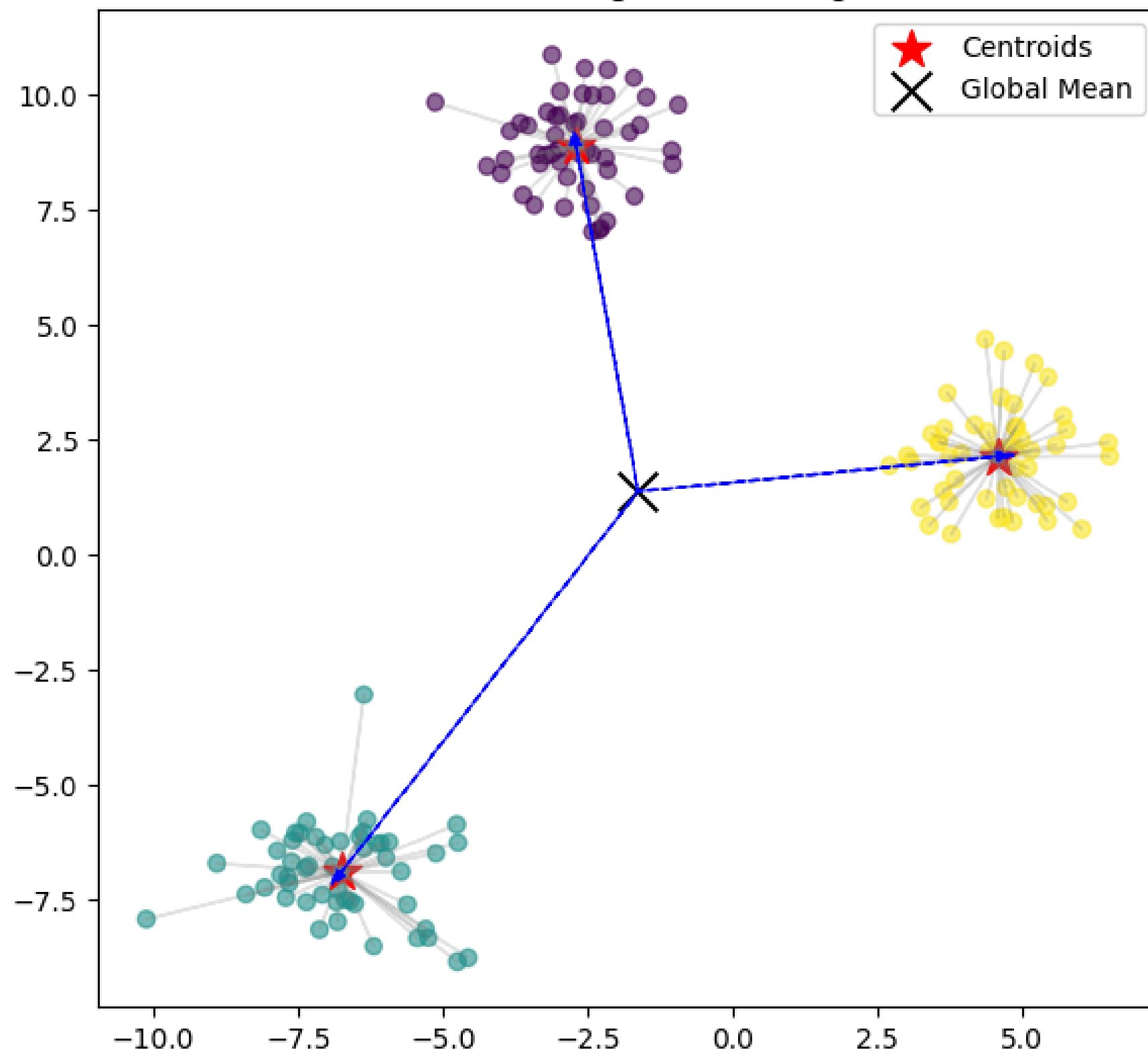
1. **WCSS** = "Are similar pieces close?"
2. **BCSS** = "Are different groups far apart?"

WCSS (Within-Cluster Sum of Squares)

BCSS (Between-Cluster Sum of Squares)

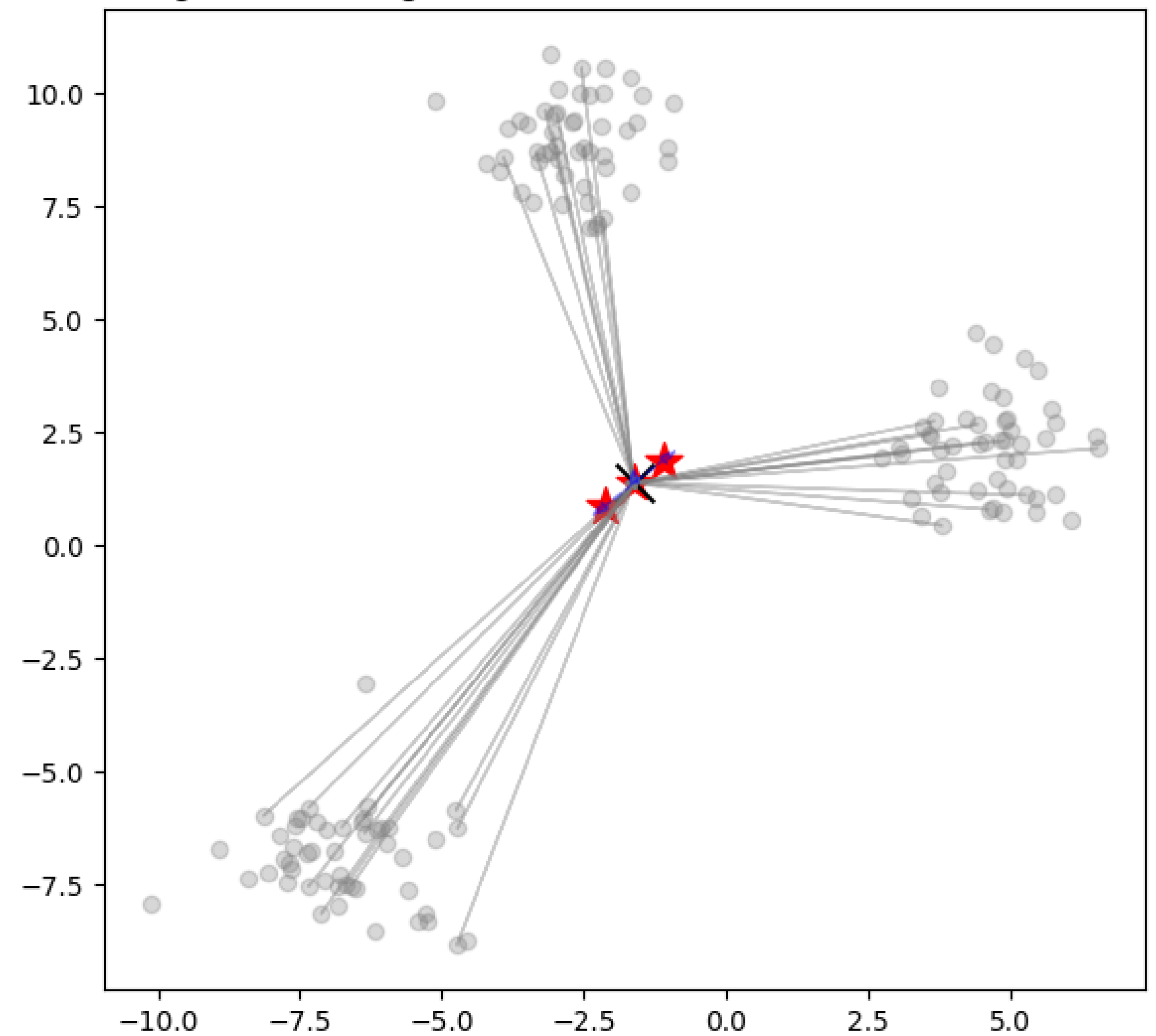
Good Clustering

Low WCSS (Short Arrows), High BCSS (Long Dashed Arrows)



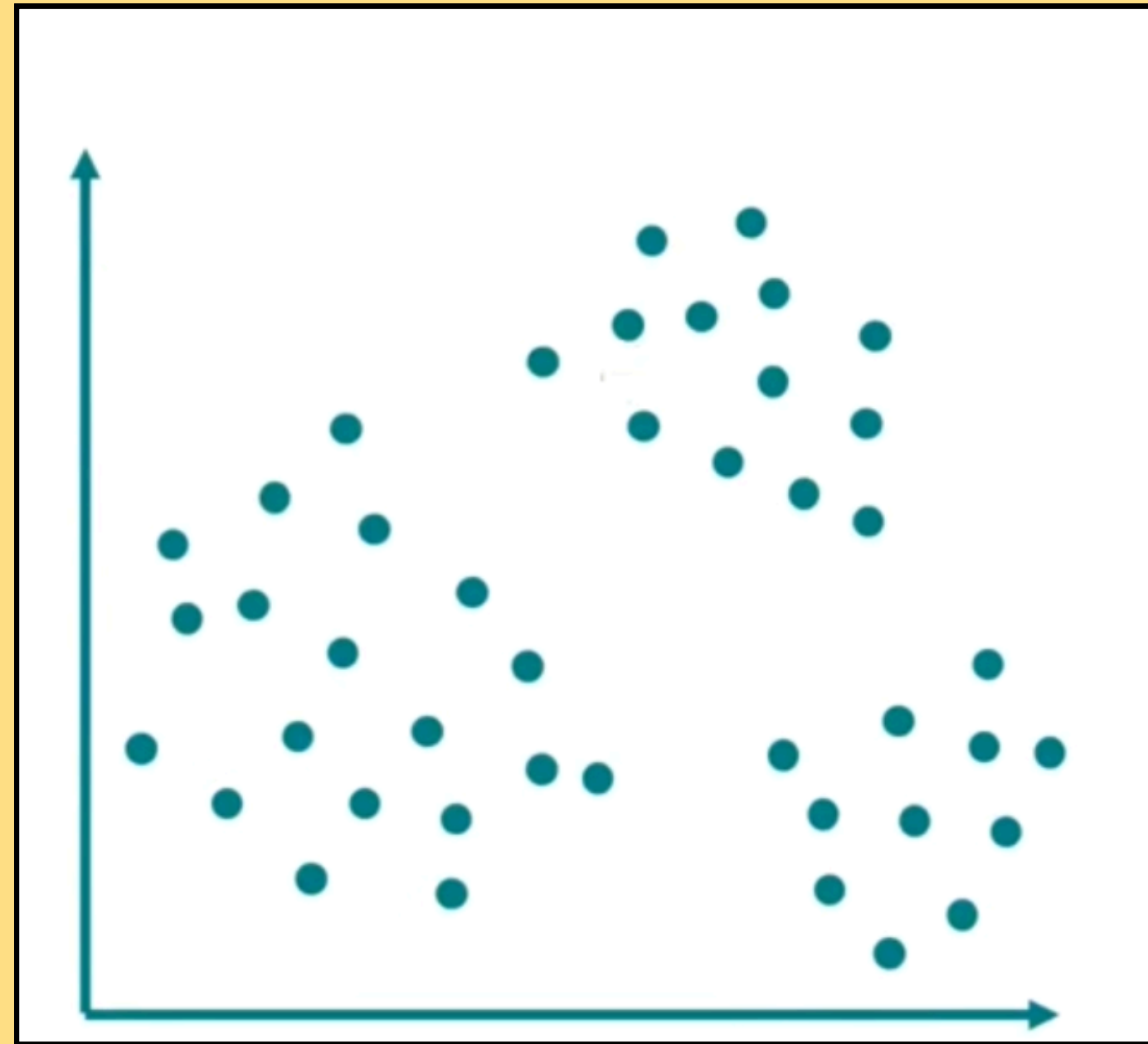
Bad Clustering

High WCSS (Long Arrows), Low BCSS (Short Dashed Arrows)

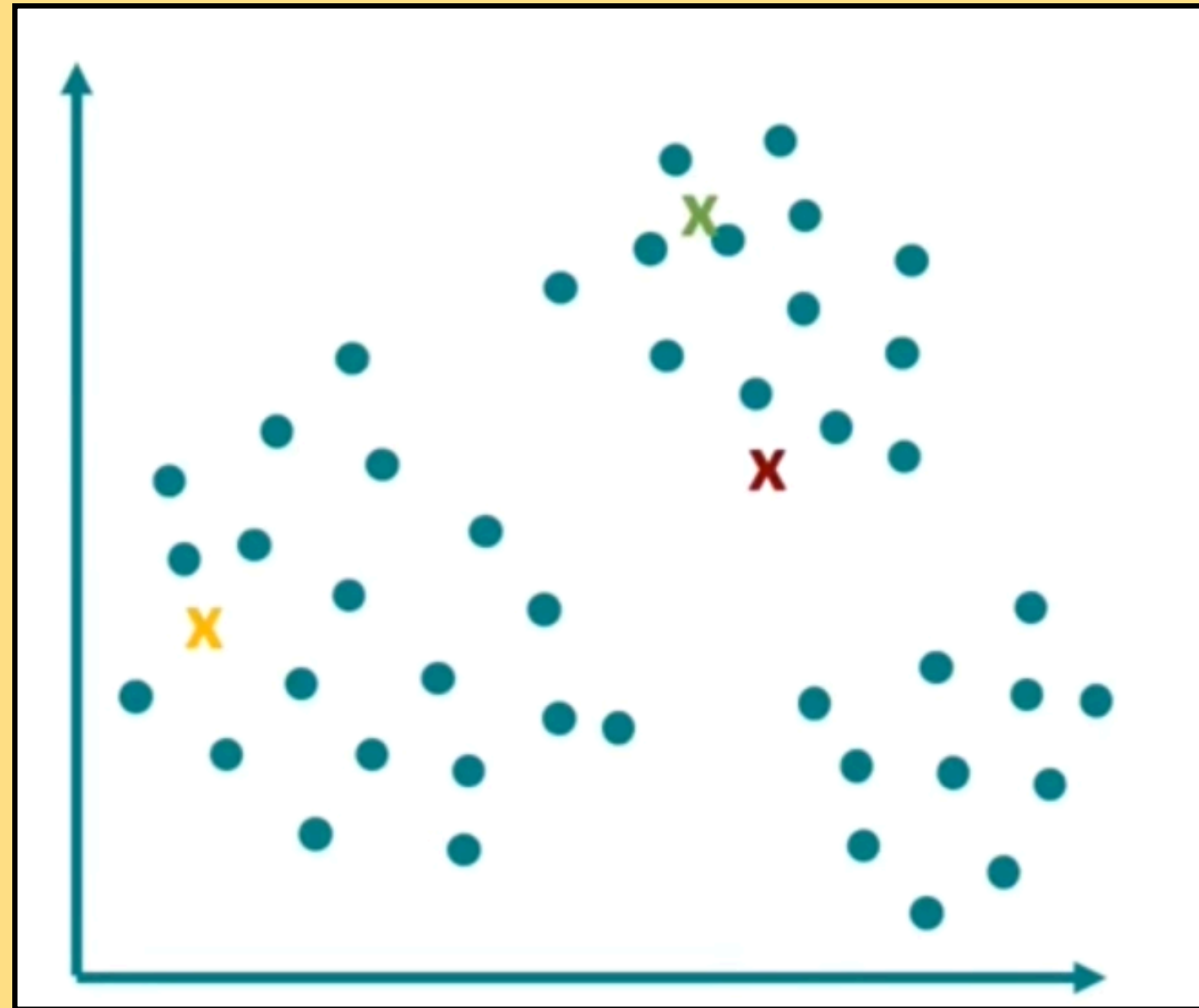


How K-means Clustering work?

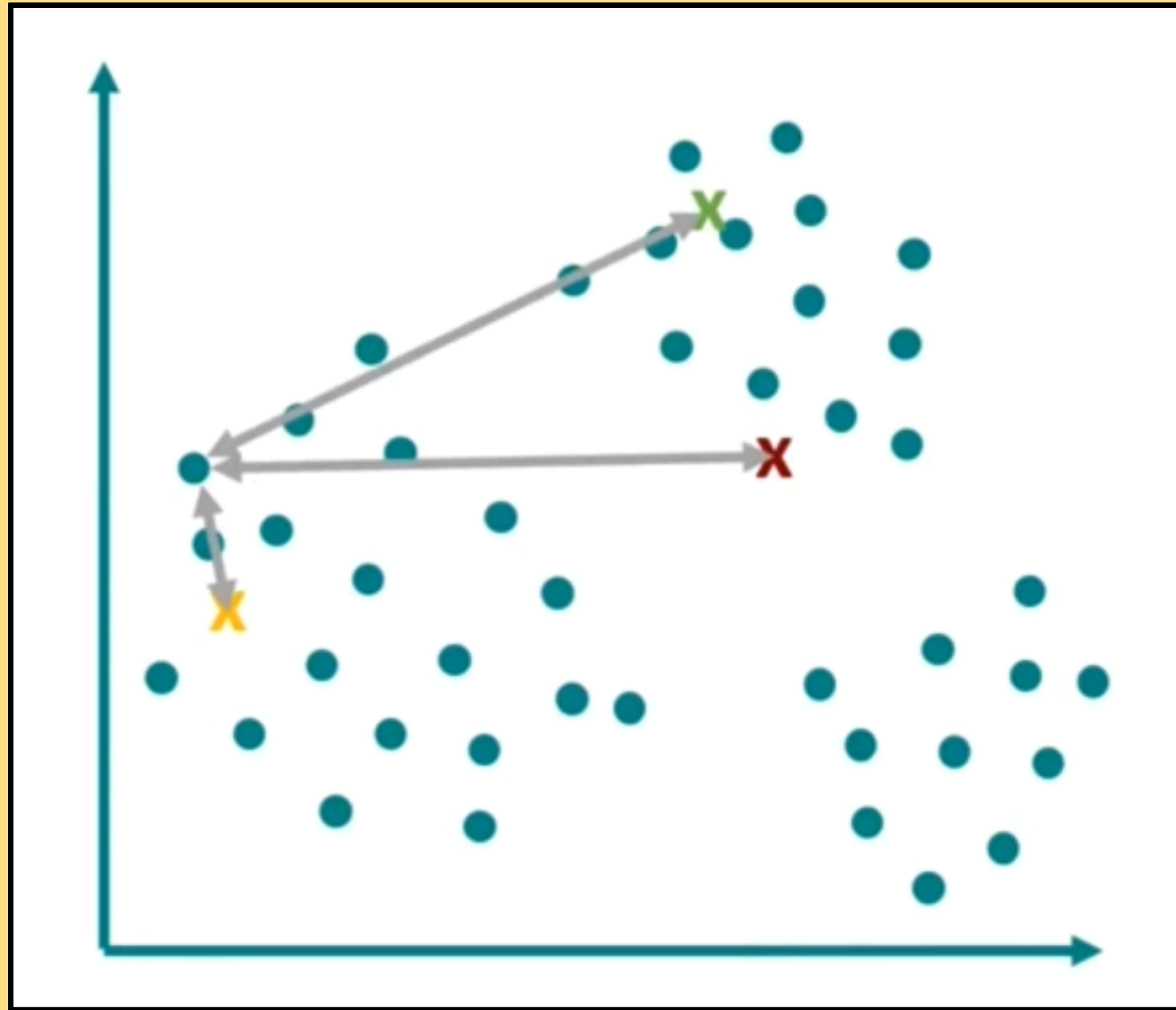
1- Define K (number of clusters) Randomly



2- Set Clusters Centers Randomly



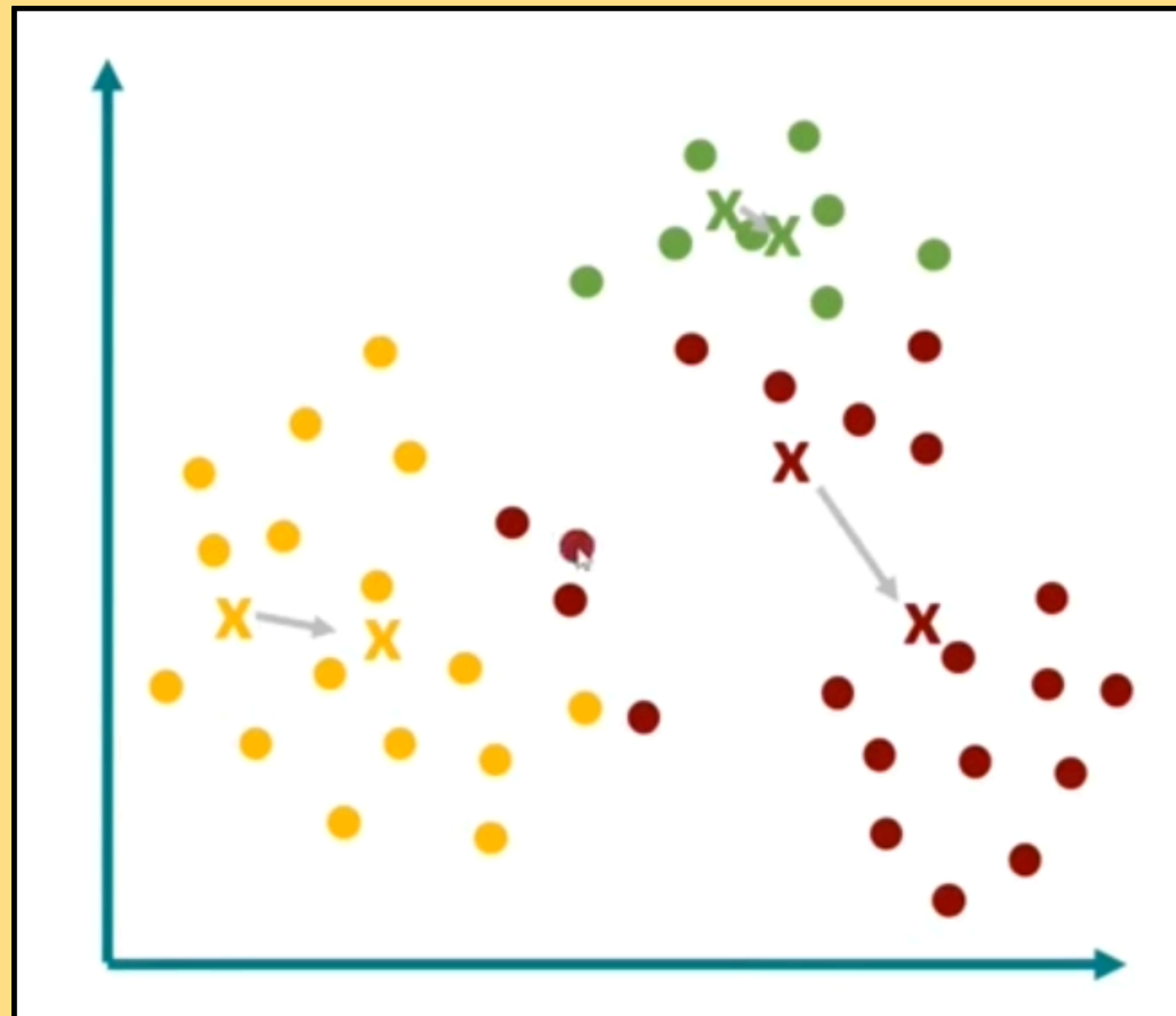
3- Assign points to clusters based on distance between the point and each center



- The most common used distance measure is “Euclidean Distance”

$$\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

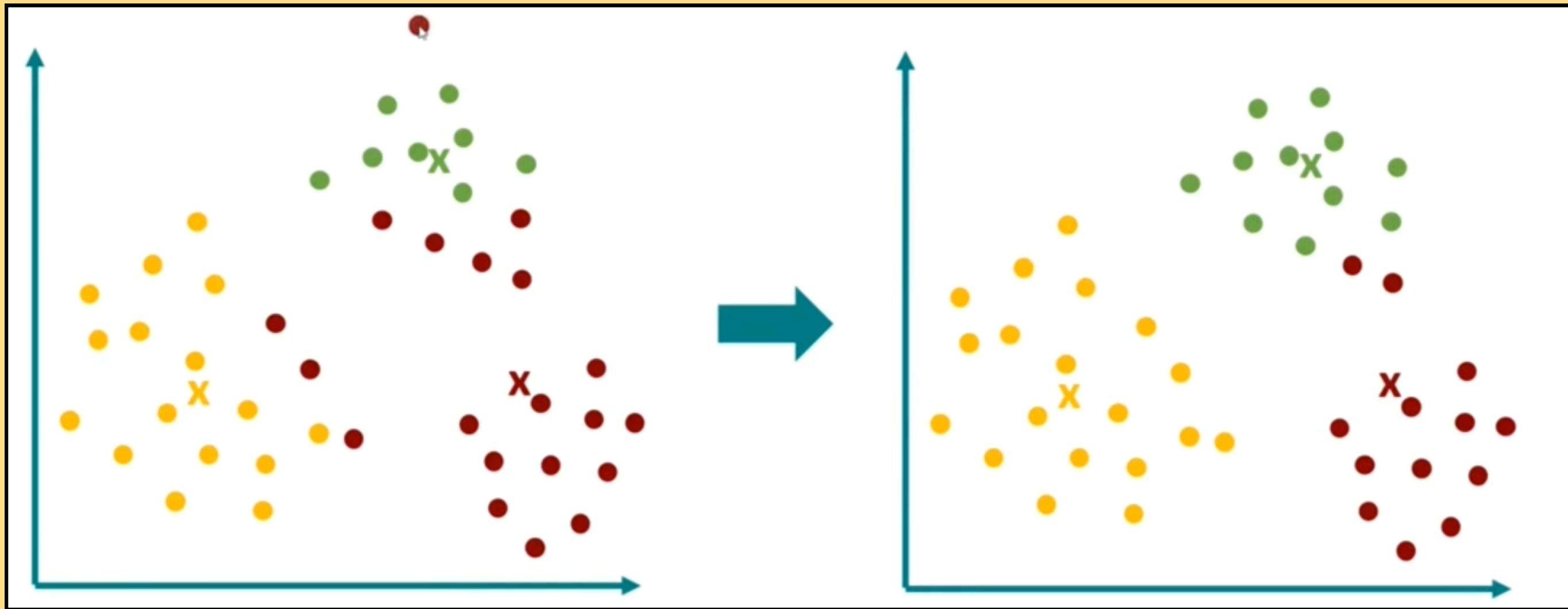
4- Updating the center of each cluster by calculating the mean.



$$\mu_k^{(t+1)} = \frac{1}{|C_k|} \sum_{\mathbf{x}_i \in C_k} \mathbf{x}_i$$

- μ_k : new centroid of the cluster
- $|C_k|$: number of points in the cluster
- $\mathbf{x}_i$: Data point assigned to the cluster

5- Assigning points to the new cluster after updating the centroids.



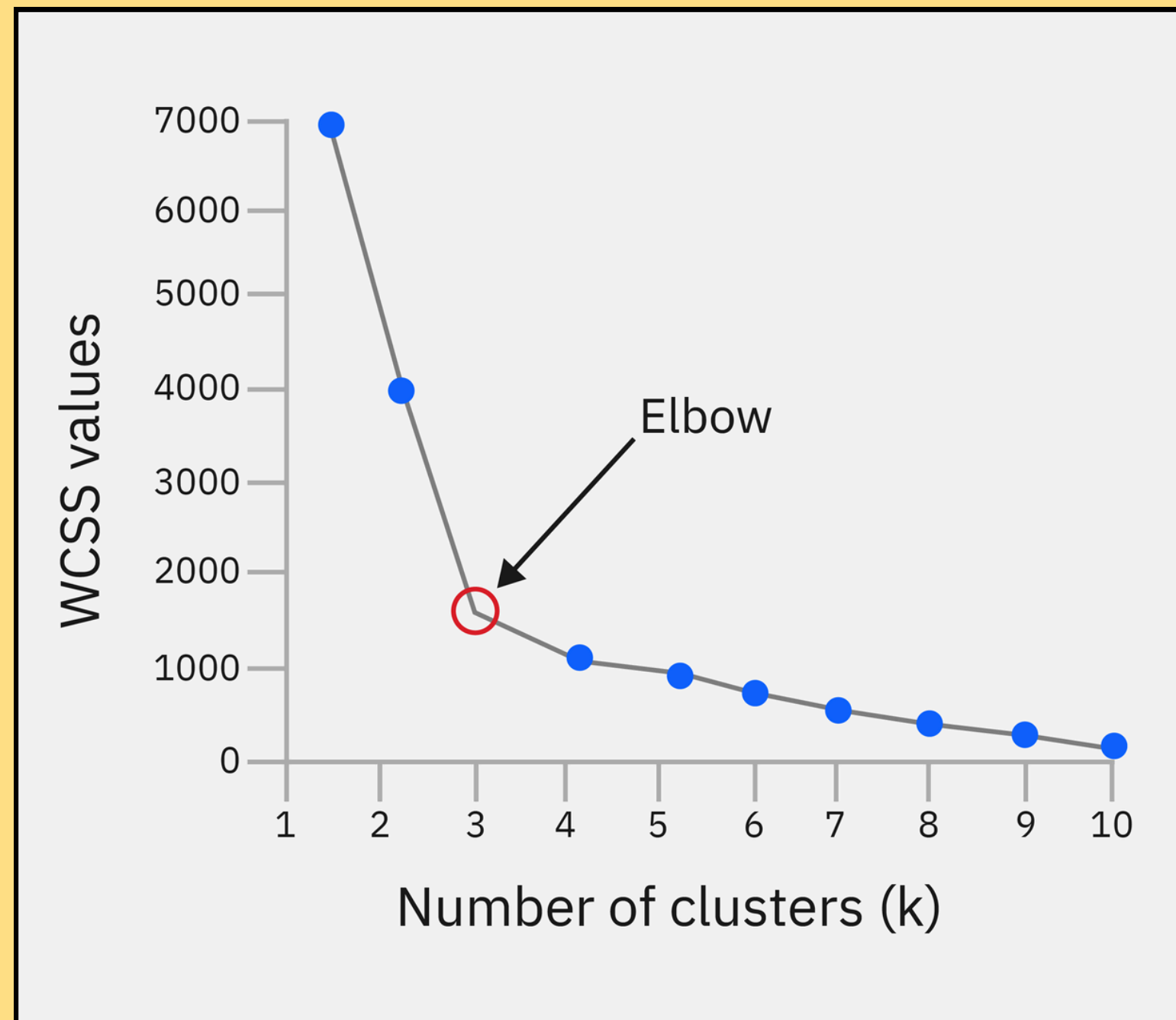
The update process is repeated until
convergence

What is “Convergence”?

- It is when the centroids no longer change significantly
- It ensures the within-cluster-variance.
- Convergence is important, but it does not mean that we found the best solution! K-means can get stuck in local optima. That’s why we run it multiple times with different initial centroids.

**Define the right
number of clusters (K)
by Elbow Method**

Elbow Method



- We iterate through a range of the possible (k) values, and for each (k) we calculate the sum of distance of each point from its cluster centroid (WCSS).
- The elbow point is the point where the (WCSS) starts to decrease at a lower rate and adding more clusters is not significantly improving the fit within the clusters.

Evaluation ??

Silhouette Score

- It measures how similar a point is to its cluster and represents it as a score that ranges **from -1 to 1** where:
 - 1 → point is well matched to its own cluster and poorly matched to neighboring clusters.
 - 0 → point is on or very close to the decision boundary between two neighboring clusters
 - -1 → point is poorly matched to its own cluster and well matched to a neighboring cluster.

$$S(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}$$

- **a(i)**: the average distance between ① and all other points in the same cluster.
- **b(i)**: the average distance between ① and all points in the nearest cluster.

Limitations of K-means

- **Dependency on the initial guess:** Final clusters can be affected by the initial guess, and the algorithm may not find the best solution leading to less accurate clusters.
- **Sensitivity to outliers:** K-means treats all data points equally and can be sensitive to outliers as it depends on the mean.
- **Need to know the number of clusters:** We have to tell the algorithm how many clusters we expect in the data.
- **K-means assumes clusters are round or spherical and equally-sized which is not always the case in the real-world.**

Tips and Considerations on K-means

Case	Recommended Solution
Categorical data	1- Consider using K-modes instead of encoding and see which is better!
Dependency of initial guess of centroids	1-Choose different random starting points in order to get the best possible result. 2- Use K-means++ which selects the first centroid randomly, then select the rest of the centroids based on the distance from the first one (the points that are so far have high probability to be chosen)

Case	Recommended Solution
Sensitivity to outliers	1- Consider using K-Medians instead of K-means to reduce the effect of the outliers. 2- Assign weights to data points, so that the far data points have a lower weights.
Too large dataset	Consider using mini-batch K-means that takes small random mini-batches of data instead of using the full dataset.

When to use K-means Clustering?

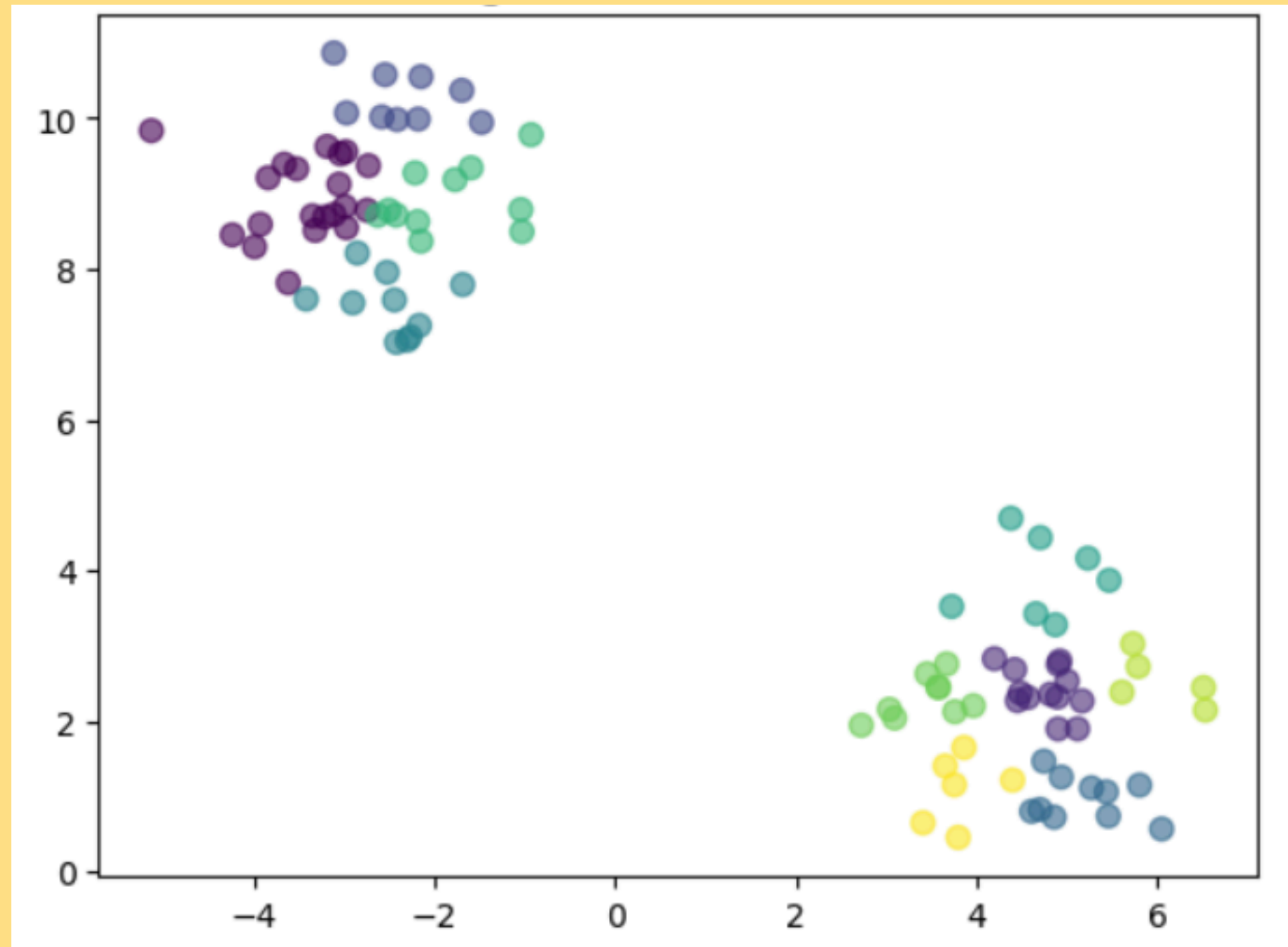
- **You have a large dataset**
- **You know (or can guess) the number of clusters (K).**
- **Clusters are roughly spherical and equally sized.**
- **Data is numeric and continuous.**
- **Suitable for quick clustering tasks**

Real-world Applications of K-means

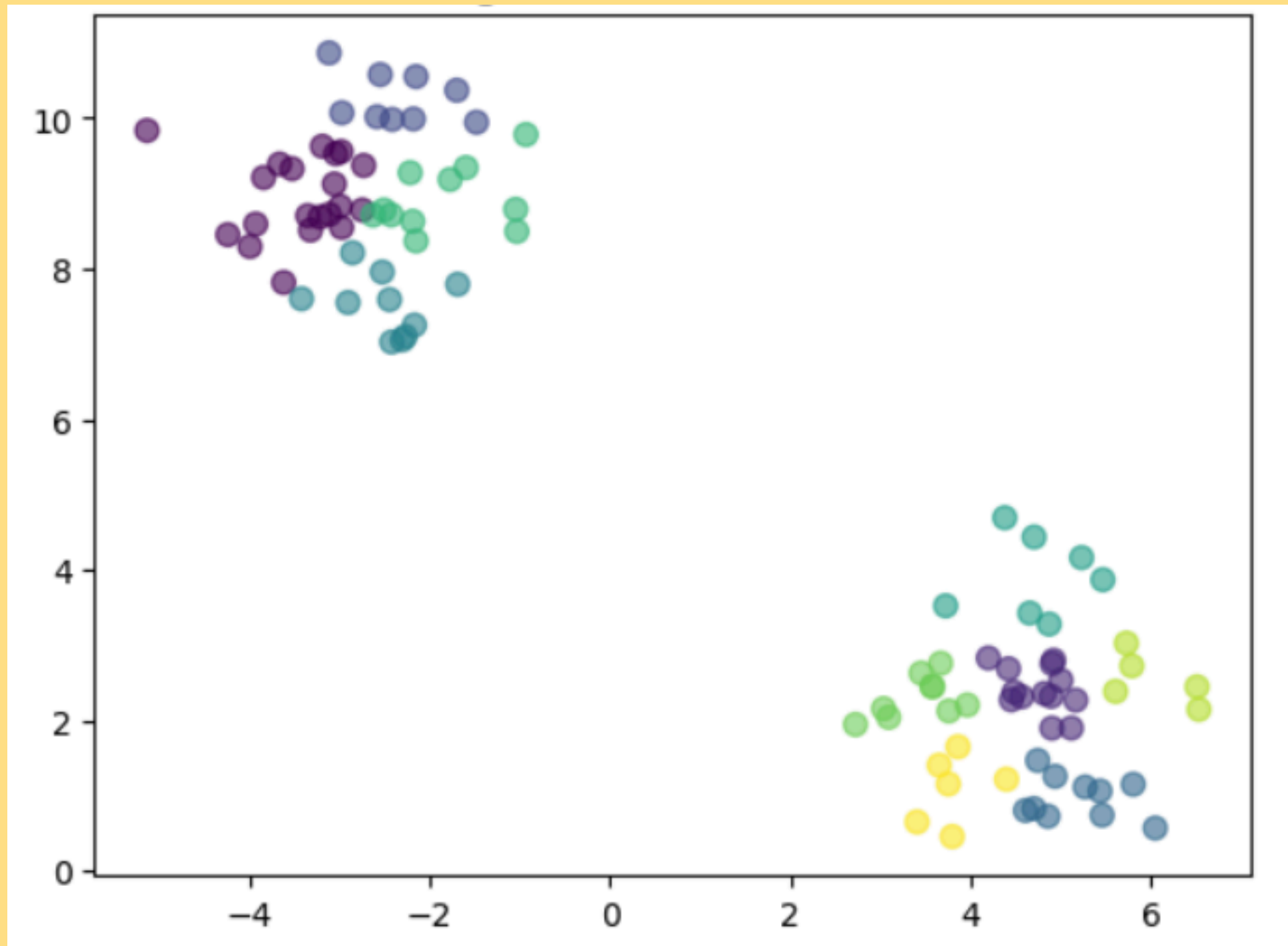
- **Recommendation Systems** K-Means is used to cluster similar products or services together and recommend them to users based on their past behavior.
- **Text Clustering** K-Means is used in natural language processing (NLP) to cluster similar documents or text data together, making it easier to analyze large volumes of text data.
- **Anomaly Detection** We assign a threshold that if a distance from a point to the center exceeds it, it is detected as an outlier or anomaly.

Activity!

Why this is a poor clustering?

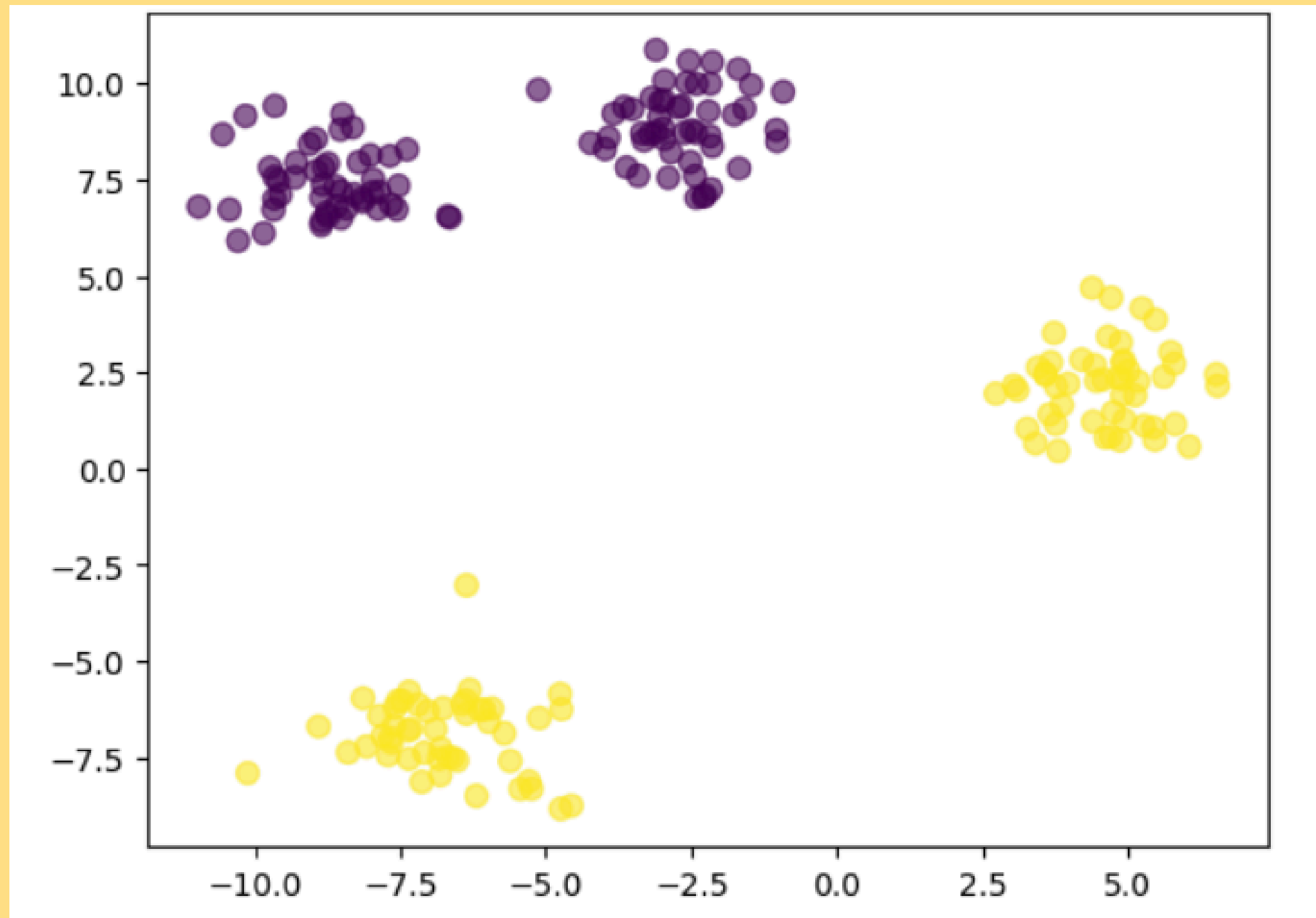


Answer:

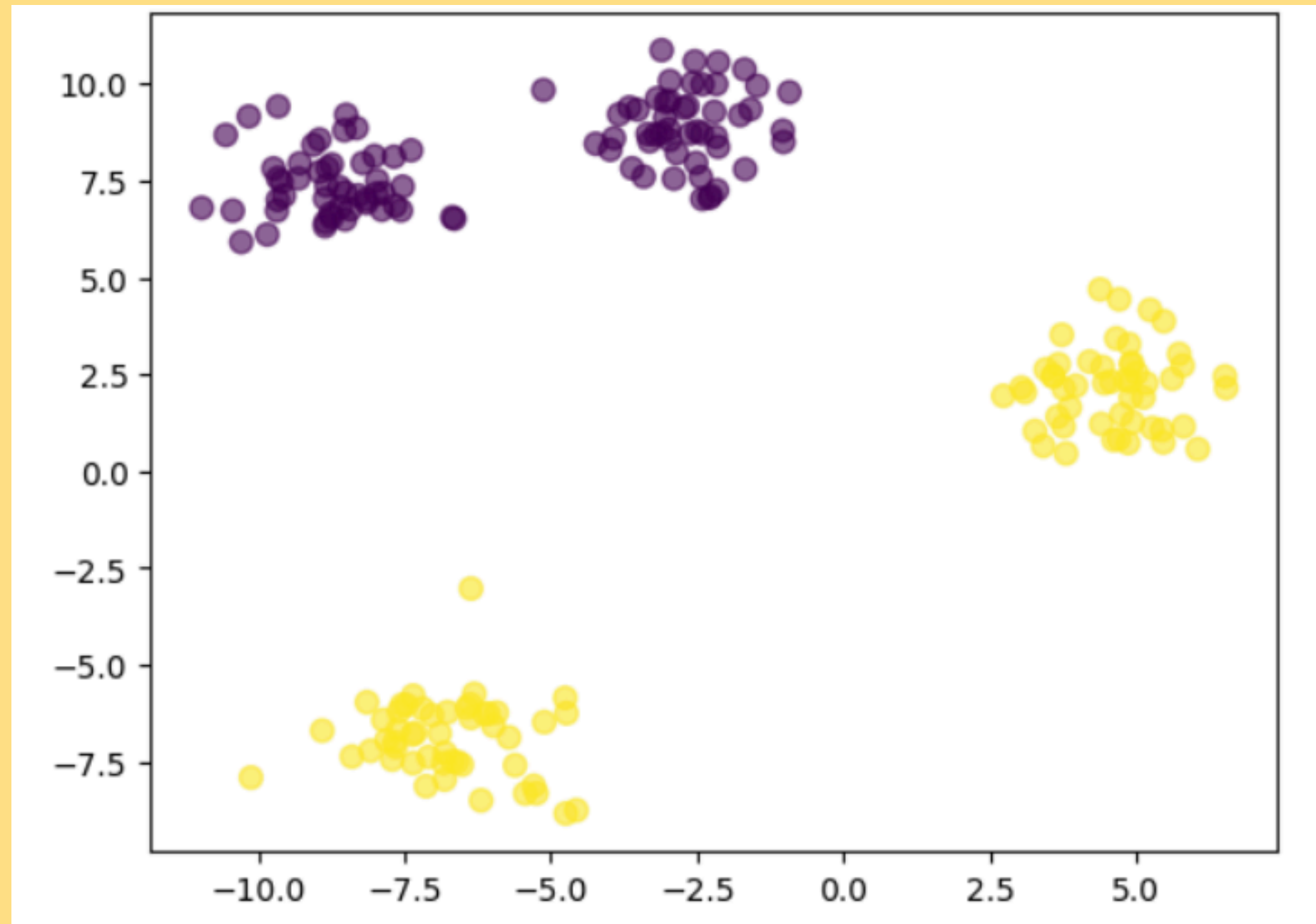


K is not right:
K number is **too large**, so
the model is **overfitting**
(giving us more
clusters than we
really need)

Why this is a poor clustering?

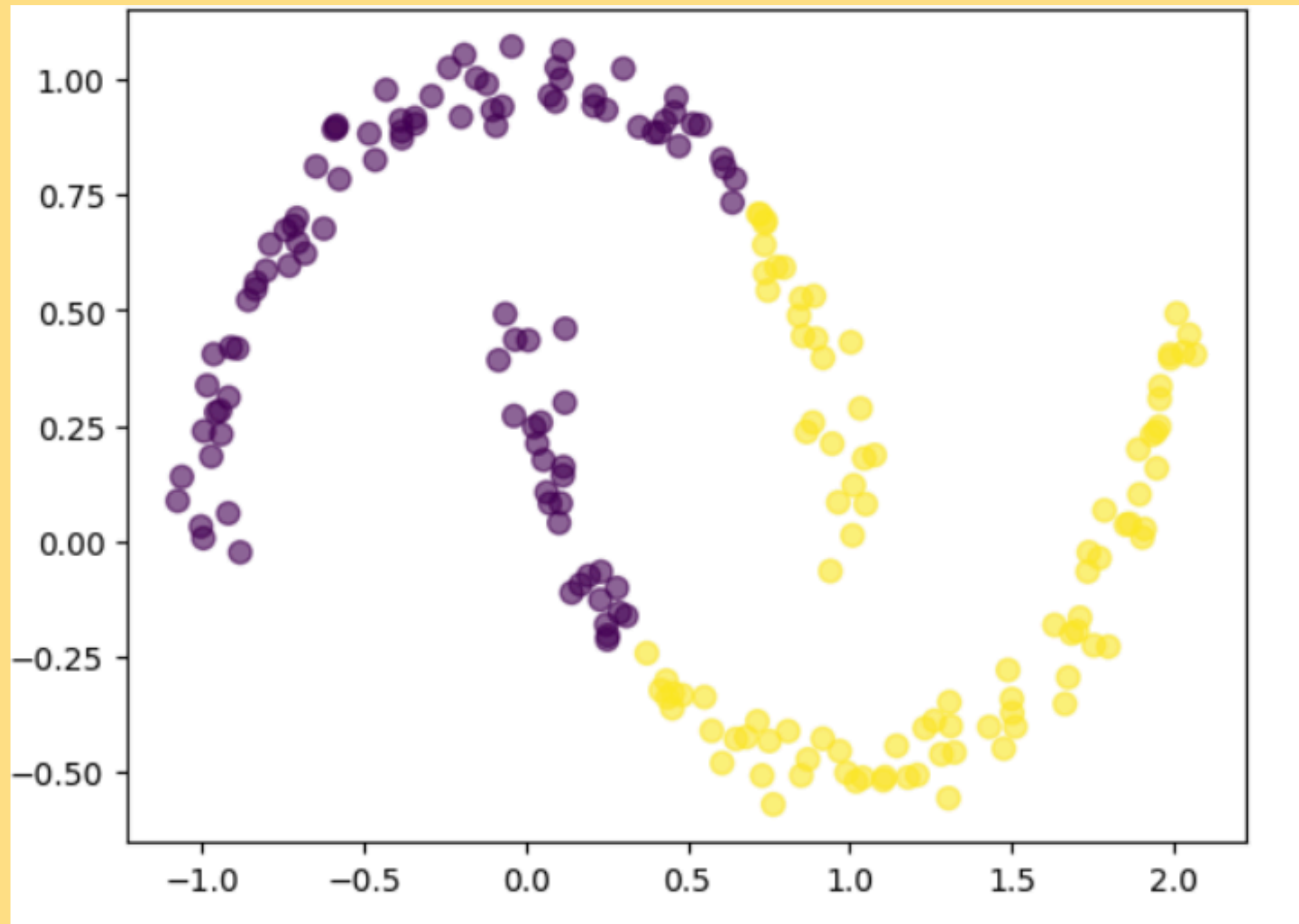


Answer:

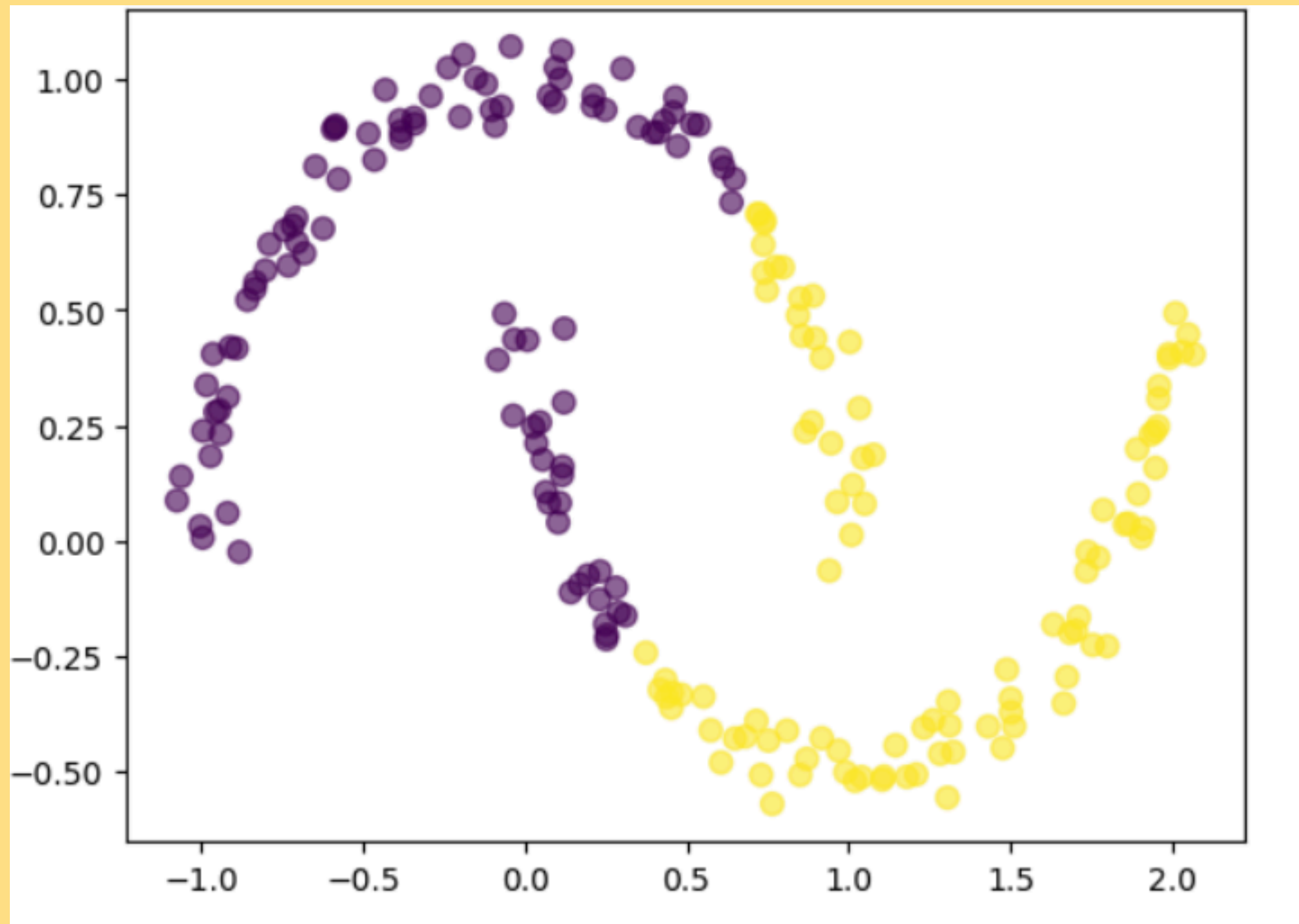


K is not right:
K is **too few**, so the model is **underfitting** (giving general clusters that are not useful and less than what is supposed to be)

Why this is a poor clustering? What is the suitable algorithm for this problem?

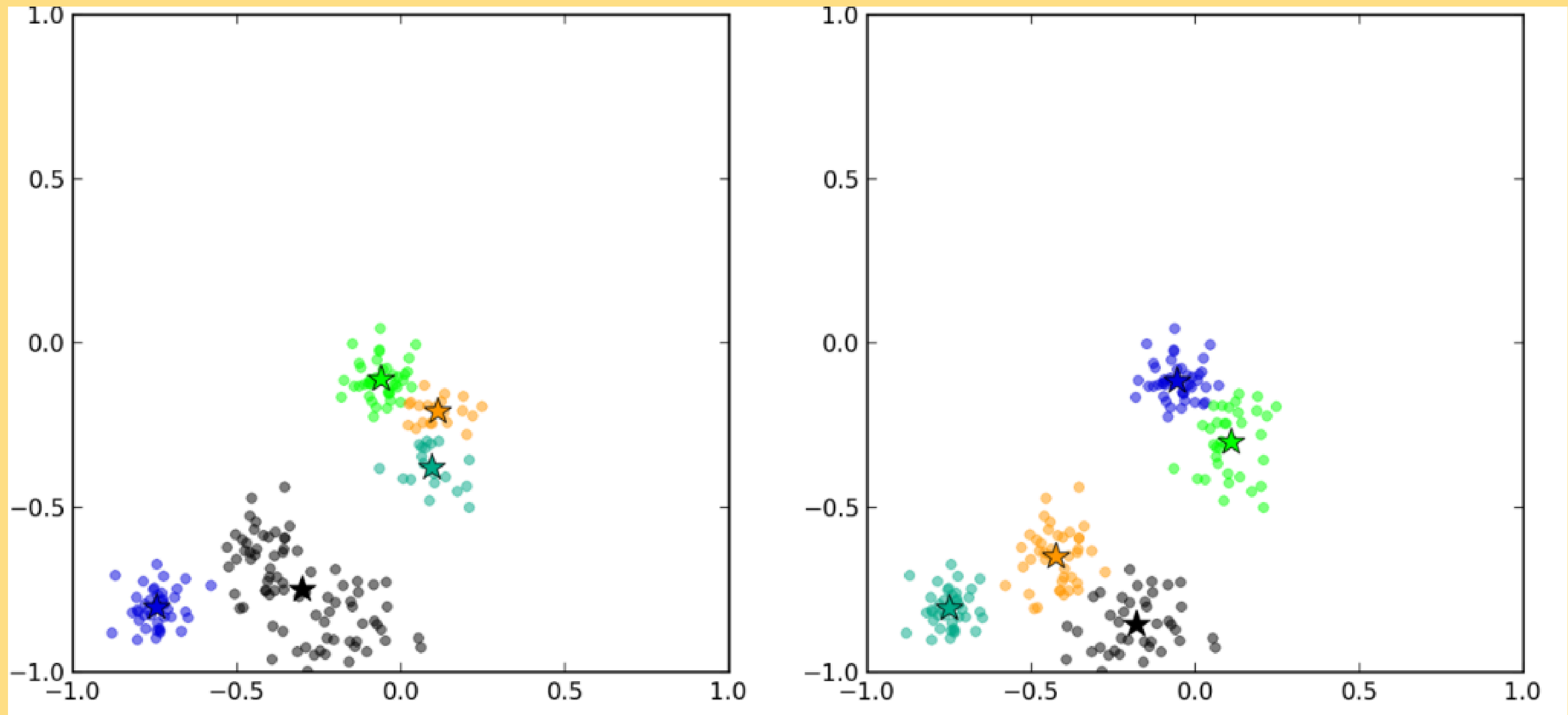


Answer:



Clusters are not spherical, so the model failed to cluster them correctly. We can use another algorithm such as hierarchical or DBSCAN.

What is the best clustering? and guess why it is better.



Answer:

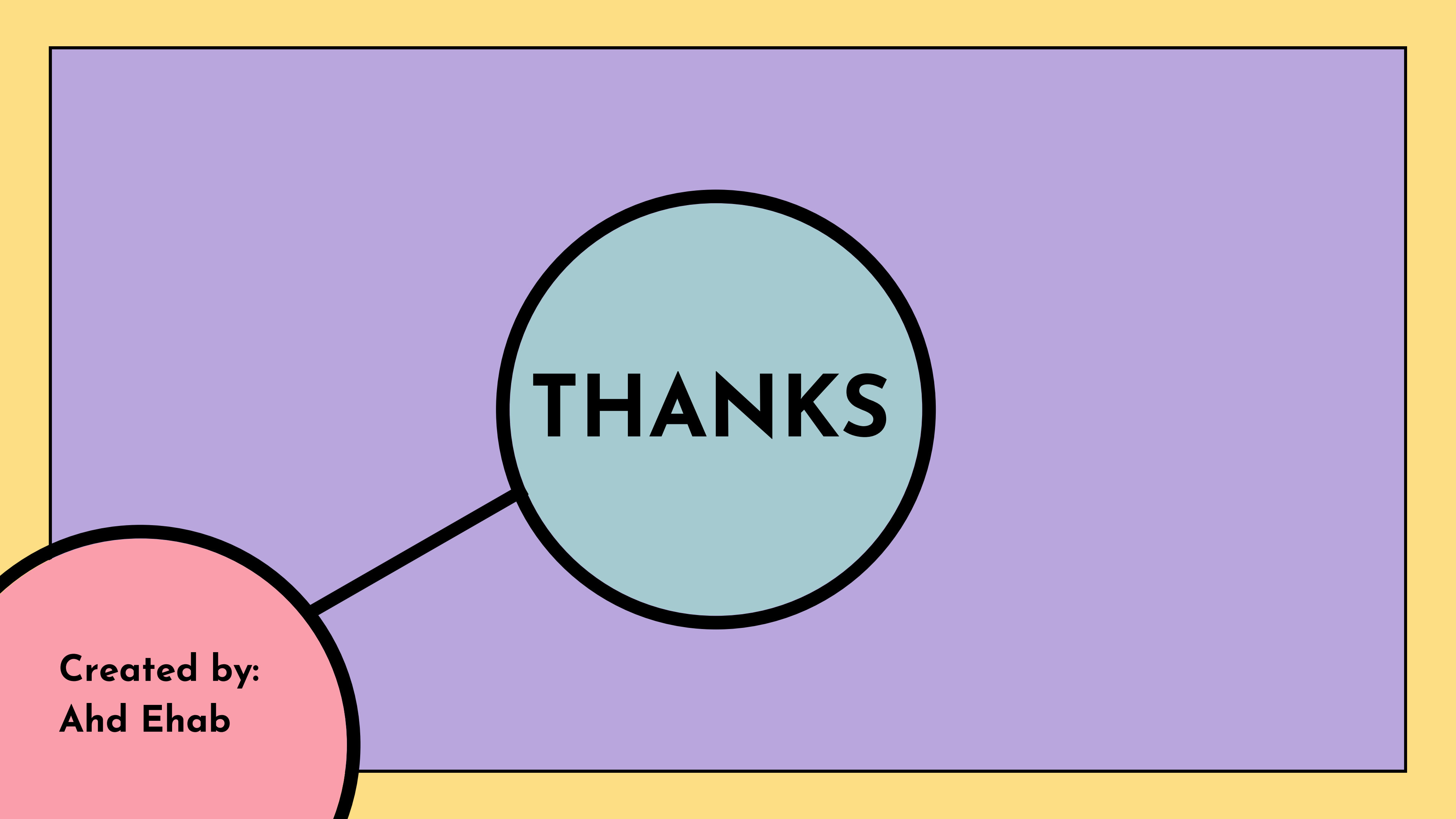
Left Side:

Random initial guess of k gives us poor clustering, the centroids will reach convergence, but will never reach the optimal solution

Right Side:

The clustering result is much better because we used **K-means++** instead of random initial guesses.

Any Questions?



THANKS

**Created by:
Ahd Ehab**